



Ujwala Senapathi

Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [LinkedIn](#)

Summary:

- Around 5 years of Experienced as a **Data Engineer & Analyst** with expertise in designing and optimizing data pipelines, integrating real-time and batch processing workflows, and enhancing business intelligence capabilities.
- Strong background in **big data platforms** such as **Apache Spark** (DataFrames, Spark SQL, Spark Streaming), **Apache Kafka**, and **AWS** (EC2, S3, Redshift, Lambda, Glue, Kinesis) for processing large-scale datasets.
- Azure **Databricks, Spark SQL**.
- Knowledge in retrieve, analyze and present data using **Azure Data Explorer/Kusto**.
- Extensive experience in developing **ETL workflows** using **Apache Airflow**, automating data transformations, and orchestrating data movement between systems.
- Proficient in **real-time data streaming** solutions, using **AWS Kinesis** and **Kafka**, processing millions of events per second with **Spark Streaming**.
- Expertise in **containerized environments**, deploying and managing applications with **Docker** and **Kubernetes** for efficient resource management and scalability.
- Skilled in **data warehousing** and **cloud-native architectures**, with a focus on **AWS Lambda** (serverless execution), **AWS Glue** (scalable ETL), and **AWS EMR** (distributed Spark workloads).
- Hands-on experience with **databases** such as **Teradata, MySQL, MongoDB, and Cassandra**, optimizing queries, managing data storage, and ensuring high availability.
- **Data governance and compliance** focus, with experience ensuring data quality, consistency, and HIPAA compliance for sensitive healthcare data.
- Expertise in **query optimization**, including **Teradata EXPLAIN, Partitioned Primary Indexing (PPI)**, and advanced SQL techniques for performance tuning.
- Curate data sourced out of Lake in to **Databricks** in different phases environments and perform Delta strong data engineering experience in **Spark and Azure Databricks**, running notebooks using **ADF**.
- Proficient in **data visualization** using **Tableau** and **Python (Matplotlib, Seaborn)** for creating insightful reports and visualizations to aid decision-making.
- Implemented **CI/CD pipelines** for automating data workflows and ensuring consistent deployments using **Git, Jenkins, GitLab CI, and AWS Code Pipeline**.
- Implemented and maintained **data serialization and deserialization** pipelines using **AVRO, JSON, and Parquet** to support both real-time and batch data processing workflows.
- Experienced in **automated testing** for ETL jobs and data validation, ensuring data integrity during processing.
- Strong collaborator, working cross-functionally with business teams to define requirements, prioritize tasks, and deliver data-driven solutions.
- Collaborated with the security team to enforce **data security** practices across the data infrastructure, including the use of **KMS, IAM roles, and data encryption** during both transmission and storage.
- Adept at optimizing and automating data workflows and pipelines, ensuring scalability, high availability, and cost efficiency across cloud and on-prem environments.
- Demonstrate strong communication skills by effectively conveying complex technical concepts to diverse stakeholders.

Technical Skills:-

- **Programming Languages:** Python, Bash, PowerShell



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [Linked In](#)

- **Big Data Technologies:** Apache Spark (Data Frames, Spark SQL, Spark Streaming), Apache Kafka, Apache Cassandra
- **Cloud Platforms & Services:** AWS (EC2, S3, Redshift, Glue, Lambda, Kinesis, Step Functions, CloudWatch, EMR, ECS, EKS, Fargate, IAM roles, VPC configurations), AWS Lambda, AWS Glue, AWS Step Functions
- **Data Warehousing & ETL:** Informatica PowerCenter, DataStage, Apache Airflow, ETL Automation, Data Integration, Azure SQL
- **Data Storage & Databases:** MySQL, SQL Server, Amazon Redshift, MongoDB, Apache Cassandra, Teradata, NoSQL, Data Modelling
- **Containerization & Orchestration:** Docker, Kubernetes, Helm, AWS ECS, AWS EKS, AWS Fargate
- **Real-time Data Streaming:** Apache Kafka, AWS Kinesis, Spark Streaming
- **Data Governance & Security:** Data validation, error handling, data auditing, compliance (HIPAA for healthcare data), IAM roles, data privacy
- **CI/CD & Version Control:** Git, GitHub, GitLab CI, Jenkins, AWS Code Pipeline, Version Control Best Practices
- **Data Visualization & Analytics:** Tableau, Matplotlib, Databricks, Seaborn, Jupiter Notebooks, Excel (Pivot charts)
- **Data Formats & Transformation:** JSON, XML, CSV, Parquet
- **Query Optimization:** SQL Query Optimization, MySQL, Teradata EXPLAIN, Indexing, Buffer Tuning, Partitioning (PPI), Hash Joins
- **Job Scheduling & Automation:** Cron Jobs (Linux), Autosys, Apache Airflow, Workflow Automation Tools
- **Data Migration & Replication:** Data replication and transformation (AWS Lambda, S3, Glue, Data pipelines)
- **Cloud Data Integration & Orchestration:** AWS Glue, AWS Lambda, Apache Airflow, Apache Kafka, AWS Kinesis
- **Machine Learning & Data Processing:** PySpark, MLlib (Machine Learning in Spark), Data preprocessing
- **Data Quality & Auditing:** Ensuring data accuracy, integrity, and compliance throughout the ETL pipeline
- **Monitoring & Logging:** AWS CloudWatch, Prometheus, AWS X-Ray, error tracking
- **Linux/Unix Systems:** Ubuntu, CentOS, System Administration, Task Automation

Professional Summary :-

Wells Fargo, San Francisco, CA
Software Data Engineer

May 2024 – Present

Responsibilities:

- **Utilized AWS EMR with Spark** for performing batch processing on large-scale data sets from various Big Data sources, leveraging both **Spark Data Frames** and **Spark SQL** for high-performance querying.
- **Developed and managed AWS EC2 instances**, deploying applications in a **containerized environment** using **Docker** and **Kubernetes** for efficient resource management and scalability.
- **Built data pipelines** using **Apache Airflow**, orchestrating the extraction, transformation, and loading (ETL) of data from multiple **APIs** and flat files into **AWS S3** for further processing and analysis.
- **Deployed AWS Lambda functions** that are triggered by events in **S3** buckets to perform real-time data transformation and loading into **AWS Redshift** for analytics.
- **Integrated real-time streaming data** from **AWS Kinesis** and **Kafka** streams, loading data into **AWS S3** and processing using **Spark Streaming** for real-time analytics and decision-making.
- Developed **Python and shell scripts** to automate **Spark job orchestration** and batch processing workflows for ingestion, transformation, and analysis of data.
- Developed and implemented Historical and Incremental Loads using **Databricks & Delta Lake** run using **ADF pipelines**.
- Created pipelines to load data from Lake to Databricks and Databricks to **Azure SQL DB**.



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [Linked In](#)

- Developed a new data scheme for the data consumption store for the **Machine Learning and AI** models to quicken the processing time using **SQL, Hadoop, and Cloud services**.
- **Utilized Amazon EC2, Auto Scaling, and Elastic Load Balancing (ELB)** to configure
- Assisted in creating **data de/serialization** strategies using **Parquet** and **JSON** to optimize data storage and performance and scale compute resources based on workload demands, ensuring high availability and cost efficiency.
- **Optimized data pipelines** using **AWS Glue** for ETL operations, integrating data from Kinesis stream data into **AWS S3** and other data sources for downstream processing.
- **Extracted and analysed data** using **Apache Spark** from **AWS Redshift** and other cloud storage services to provide meaningful insights and support business analytics.
- **Implemented data quality checks** and data governance measures on incremental loads to ensure accuracy and consistency across datasets.
- Migration large data sets to **Databricks (Spark)**, create and administer cluster, load data, configure data pipelines, loading data from ADLS Gen2 to Databricks using **ADF pipelines**.
- Led the integration and transformation of data using **Data Serialization** formats (Parquet, AVRO, JSON-LD) to ensure efficient data storage and retrieval.
- Lead multiple **AI (Artificial Intelligence) and Machine Learning** programs within our product suits; Programs include - Predictive Analytics service - baselining and forecasting of performance and **security KPIs, Security Analytics - Anomaly detection service** - clustering of devices based on behaviour over time - **NLP, LSTM, KubeFlow, Docker, AWS Sagemaker, AWS Greengrass**.
- **Performed data normalization and data visualization** using **Python libraries** (e.g., Matplotlib, Seaborn) to create correlation plots and scatter plots, identifying trends and patterns in large datasets.
- **Enhanced data pipelines** with **AWS Step Functions**, building state machines to automate complex workflows, handling error retries, and scheduling pipeline runs to process data efficiently.
- **Worked with SQL databases** like **MySQL, Redshift, and MongoDB** to ensure data integrity, perform complex queries, and optimize performance for both transactional and analytical workloads.
- **Implemented data-driven decision-making** tools for business stakeholders by designing dashboards and ensuring that data presented was accurate and aligned with business requirements.
- Built end-to-end **data platforms** for real-time and batch data processing, utilizing **AWS Redshift, Athena, and Kinesis** to enable high-performance analytics and reporting.
- **Collaborated cross-functionally** with business teams to define data requirements, identify data sources, and prioritize business needs for data-driven decision-making and product development.
- **Containerized and deployed applications** with **Docker** and **Kubernetes**, enabling better scalability, easier management, and consistent environments across development, testing, and production stages.
- Regularly participate in code reviews and collaborative development to enhance team efficiency and knowledge sharing
- **Monitored and optimized data pipelines** for performance, leveraging tools such as **AWS CloudWatch** and **AWS X-Ray** for tracking errors and improving pipeline efficiency.

Environment:

Python, Bash, SQL, AWS (EC2, S3, Redshift, Glue, Lambda, Step Functions, Kinesis), Docker, Kubernetes, Apache Spark (DataFrames, Spark SQL, Spark Streaming), Apache Kafka, AWS Glue, AWS EMR, AWS Redshift, Cassandra, MongoDB, Databricks, MySQL, AWS Redshift, Apache Airflow, AWS Step Functions, CI/CD Pipelines, Python (Matplotlib, Seaborn), Jupiter Notebooks, Docker, Kubernetes, Helm, Git, CI/CD, Linux

Amazon, Dayton, OH
Software Data Engineer
Responsibilities:

Jun 2023 – Apr 2024



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [LinkedIn](#)

- Implemented **Spark with Python** and utilized **Spark DataFrames** and **Spark SQL API** for efficient data processing and querying across various structured and unstructured datasets.
- Developed **Apache Spark applications** to handle data from **RDBMS (MySQL)** and real-time streaming sources via the **Kafka-Spark** integration.
- Manage Data Storage and processing pipelines in **GCP for serving AI and ML services** in Production, development and testing **using SQL, Spark, Python and AI vm**
- Worked with unstructured data formats such as **Parquet, JSON**, and **CSV**, using **Spark** for data ingestion, transformation, and analytics.
- Streamed real-time data from **Kafka** using the **Spark Streaming API**, processing millions of events per second.
- Designed various azure data factory pipelines to pull data various data sources and load the data into **Azure SQL database**.
- Led the design and development of **Data Frameworks** that streamlined ETL workflows using **Apache Spark, AWS Glue**, and **Python**, significantly improving processing speed and data quality.
- Consumed **Kafka messages** and ingested data into **Cassandra** clusters deployed in containers to support high availability and scalability.
- Developed preprocessing jobs to flatten complex **JSON files** into structured formats using **Spark DataFrames**.
- Ingested and transformed data from **MySQL** into **Cassandra**, enabling efficient data storage for large-scale applications.
- Assisted in **Cassandra schema design**, including **keyspace creation, table creation**, and **secondary index creation** to ensure optimal query performance.
- Optimized **Cassandra queries** using load testing and **Cassandra stress tool** to fine-tune performance under high-volume scenarios.
- Created **proof-of-concept (POC) applications** using **PySpark** and **MLlib** to demonstrate machine learning capabilities within the data pipeline.
- Databricks job configuration, Refactoring of **ETL Databricks notebooks**.
- Collaborated with cross-functional teams to deploy and manage **Cassandra clusters** using **Kubernetes** in a **Linux** environment, ensuring scalability, high availability, and resource optimization.
- Gained expertise in **Kubernetes architecture**, managing **Pods, nodes**, and **services** using **kubectl** and **Helm** for easier deployment and management of containers.
- Designed and deployed **Docker images** for applications running in **Kubernetes clusters** to ensure a consistent and efficient environment across development, testing, and production stages.
- Assisted in the **schema design and optimization** of **Cassandra** and **MongoDB** to ensure high performance and fast retrieval for front-end applications.
- Managed the **MongoDB environment**, focusing on aspects such as **replication, sharding, load balancing**, and **indexing** to ensure high availability and performance.
- Worked with **GitHub** for version control, managing feature branches, pull requests, and code reviews as part of the **CI/CD pipeline**.
- Collaborated in **CI/CD pipeline integration** for continuous deployment of data processing and analytics services.
- Created pipelines to load data from **Lake to Databricks** and **Databricks to Azure SQL DB**.
- Worked in a **UNIX/Linux** environment, leveraging the command line for system administration and task automation.
- Used **Apache Airflow** for automating complex data workflows and job scheduling.
- Integrated **AWS services** for deployment and management, utilizing **Amazon EMR** for distributed Spark workloads and **AWS Lambda** for serverless execution.



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [LinkedIn](#)

- Assisted in building **data pipelines** for ingesting, processing, and storing large-scale datasets in **cloud environments** using **AWS S3, Glue, and Lambda**.
- Gained experience working with **AWS ECS, EKS, and AWS Fargate** for containerized services and serverless **Kubernetes** management.
- Managed **IAM roles, VPC configurations**, and followed **AWS security best practices** for managing access control and ensuring data privacy.
- Leveraged **AWS CloudWatch** and **Prometheus** for monitoring, logging, and alerting on critical data pipelines and containers.
- Built automated data pipelines using **Apache Airflow** and **AWS Step Functions**, enabling robust error-handling workflows and retry logic.
- Processed **real-time data streams** and implemented **event-driven architectures** using **Apache Kafka** and **AWS Kinesis**.
- Created data visualizations using **Python libraries** such as **Matplotlib** and **Seaborn** to provide actionable insights.
- Developed **cloud-native applications** and microservices using **Docker** and **Kubernetes**, optimizing resource management for containerized workloads.
- Promote continuous improvement by advocating best practices, optimizing workflows, and enhancing code quality.

Environment:

Python, Bash, SQL, Apache Spark, Apache Kafka, Apache Cassandra, AWS EMR, AWS (S3, EC2, Lambda, Kinesis, Redshift, Step Functions, EMR), Docker, Kubernetes, Helm, Amazon EKS, AWS Fargate, Cassandra, MongoDB, MySQL, GitHub, Jenkins, AWS CodePipeline, PySpark, Spark SQL, Spark Streaming, AWS CloudWatch, Prometheus, Git, GitHub, Linux (Ubuntu, CentOS), UNIX

Legato Health Technologies (Anthem), India

Sep 2020 – Aug 2022

Data Engineer

Responsibilities:

- Developed **Python scripts** for **ETL automation** to extract, transform, and load data into **Teradata** data warehouse.
- Utilized **Informatica PowerCenter** and other ETL tools to build complex data workflows, including transformations, lookups, routers, filters, aggregators, and update strategies for large volumes of healthcare data.
- Collaborated with **Business Analysts** to gather requirements, analyse business needs, and develop data integration solutions that met reporting and analytical needs.
- Involved in **Unit Testing** and **Integration Testing** of **Informatica Sessions** and **Batches**, ensuring proper execution and data integrity during ETL processes.
- Utilized **Autosys** and **Informatica Workflow Manager** for job scheduling, automating workflows, and managing ETL jobs for large-scale data processing.
- Developed and maintained **Informatica workflows and sessions**, ensuring data was processed and loaded into the data warehouse accurately and on time.
- Implemented **slowly changing dimensions (SCD)** type 1 and type 2 in **Informatica** to manage historical data changes in dimension tables.
- Worked on **Teradata Query Optimization**, leveraging tools like **Teradata EXPLAIN** for query optimization to enhance performance of large data sets and complex queries.
- Developed stored procedures to populate customer data with **transaction data** and **monthly summary data** into the data warehouse.



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [LinkedIn](#)

- Managed complex data load processes, ensuring successful completion of jobs and handling **error management** and **data validation**.
- Performed **data analysis** for identifying data issues and providing solutions to business users, assisting in **data validation** and troubleshooting.
- Optimized **SQL queries** for performance, including creating **indexes**, managing **partitioned tables**, and using **Teradata-specific techniques** (e.g., **PPI**, **hash joins**).
- Worked with **Linux** for job scheduling and automation, utilizing **cron jobs** and **Bash shell scripting** to manage and automate ETL tasks.
- Provided **data governance** to ensure accurate data loading and transformation, maintaining data quality and consistency throughout the ETL process.
- Gained experience with **cloud technologies** (AWS) for data migration and cloud-based ETL processes, including leveraging **Amazon S3** for storage and **AWS Lambda** for serverless ETL execution.
- Worked with **containerization and orchestration** tools like **Docker** and **Kubernetes** to deploy and manage ETL processes in containerized environments, ensuring scalability and high availability.
- Applied **data governance** and **compliance** best practices, ensuring that sensitive healthcare data adhered to regulatory standards such as **HIPAA**.
- Enhanced knowledge of **advanced SQL** techniques for optimizing large-scale data environments, focusing on **Teradata EXPLAIN**, **Partitioned Primary Indexing (PPI)**, **hash joins**, and managing **data skew**.
- Integrated **Git** for **version control** of Python scripts and ETL automation, collaborating with teams and ensuring efficient development practices.
- Developed and implemented **automated testing** strategies for **ETL jobs**, including **error handling** and validation steps to ensure the integrity of data during processing.
- Focused on **data quality** and **auditing** practices, ensuring accurate, compliant, and high-quality data through the entire ETL pipeline.
- Contributed to **CI/CD pipeline integration** for automating the deployment and integration of ETL scripts and data processing workflows, ensuring streamlined and reliable data updates.

Environment:

Informatica PowerCenter, Autosys, Cron Jobs, Teradata, SQL, MySQL, Python, Bash, Shell scripting, XML, Flat Files, Linux, Teradata, Data Warehousing Concepts (SCD, Fact, Dimension tables, Partitioning), Teradata EXPLAIN, Indexing, Hash Joins, PPI (Partitioned Primary Index), Git (for managing Python scripts and ETL automation), DataStage (for reference, if used in additional parts of the project), Autosys, Cron, Workflow Automation Tools, Data validation, error handling, and data auditing.

IBM Pvt Ltd, India

Jan 2019 – Aug 2020

Data Analyst

Responsibilities:

- Created **data visualizations** and developed **dashboards**, **stories** in **Tableau** to communicate insights from business data.
- Utilized various **Tableau charts** like **scatterplots**, **bar charts**, **pie charts**, and **heatmaps** to analyze trends and key performance indicators (KPIs).
- Analyzed **KPI results** regularly to assist in the development of performance improvement strategies for business stakeholders.
- Used **Excel pivot charts** and **VBA** to create **customized reports** and automate repetitive tasks for better analysis.
- Assisted with **query optimization**, **buffer tuning**, **backup and recovery**, installations, upgrades, and **security** for databases as part of the **profiling plan** for data processes.



Ujwala Senapathi
Data Engineer & Analyst

Email: s.ujwalaa7@gmail.com - Contact: - (937) 409-2002 - [LinkedIn](#)

- Ensured **production data** was accurately replicated into the **data warehouse** without data anomalies from the processing databases.
- Designed **databases** ensuring **referential integrity** and was involved in **logical design planning** for new data models.
- Analyzed **SQL queries** to improve **query optimization**, ensuring that tables used appropriate **indexes** for faster query performance.
- Created, tested, and optimized **MySQL scripts**, including **stored procedures, triggers, views**, and reports to enhance the **Data Warehouse** functionality.
- Troubleshoot and fine-tuned **databases** for performance and concurrency, ensuring the system operated efficiently under high loads.
- Automated the **code release process**, reducing the total release time from **8 hours** to **1 hour** by implementing **continuous integration (CI)**.
- Developed a fully automated **CI/CD pipeline** using **Git, MySQL**, and custom tools developed in **Python** and **Bash**, streamlining deployment and testing workflows.
- Worked extensively with **Tableau Desktop** to generate insightful reports, writing complex **SQL queries** to extract data directly from databases.
- Played a key role in the **transition** from **Subversion (SVN)** to **Git**, resulting in enhanced collaboration, version control, and efficiency for the development team.
- Collaborated with cross-functional teams to ensure the **quality** and **accuracy** of data used in **reporting** and **analytics**.

Environment

:-

Tableau, Excel Pivot Charts, VBA, MySQL, SQL Server, Data Warehousing Concepts, Referential Integrity, Python, Bash, MySQL Scripting, Git, Subversion Jenkins, GitLab CI, Custom Python & Bash Scripts for Automation, SQL Query Optimization, Indexing, Buffer Tuning, Concurrency, Backup and Recovery, Custom ETL Processes, Data Migration, Data Replication, Linux, GitHub, GitLab